

Yanze Wang

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EXPERIENCE

Education

Ph.D. in Pure Mathematics (in progress)

Expected June 2027

Academy of Mathematics and Systems Science (AMSS), Chinese Academy of Sciences

Research interests: birational geometry — Sarkisov program, algebraically integrable foliations

M.S. in Mathematics

Academy of Mathematics and Systems Science (AMSS), Chinese Academy of Sciences

B.S. in Mathematics

Beihang University

Teaching assistant for Linear Algebra and Algebraic Geometry during undergraduate studies.

Honors & Awards

- MCM/ICM (Mathematical Contest in Modeling) — Meritorious Winner (twice)
- Hua Luogeng Scholarship, Beihang University (twice)

SKILLS

- **Mathematics:** birational geometry and the minimal model program: Sarkisov program, algebraically integrable foliations; moduli spaces.
- **AI Practice:** local PyTorch / libtorch model-training mini-projects; agent-framework building.
- **AI4Math:** human verification of AI-generated mathematical results; research harnesses and Lean 4 formalization.
- **Engineering:** torch (Python & C++); Rust, C, LaTeX; Linux and Git; vibe-coding personal CLI / TUI programs (Rust).

PROJECTS

AI Practice

- **Model training:** learned from llm-from-scratch and other tutorials — Transformer internals and training loops; reproducing small-parameter training: **Chinese GPT**, **RL**, and LoRA fine-tuning.
- **AI4Math:** learning and practicing workflows of agent architectures such as co-mathematician; adapting Rethlas to my own workflow. Formalization practice — reproducing **toric-lean** and others; contributing to the Lean 4 platform prove2.me.
- **AiKit** — agent toolkit for coding agents (pi, opencode, codex, claude code, etc.): skills / plugins, used for math research and SDD programming.

Programming

- **suckless** — personal fork of dwm / dwmblocks (C), patched and maintained by hand, no AI assistance.
- **Calman** — terminal task & event manager (CLI, Rust). CalDAV-compatible; sync delegated to radicale / vdirsyncer / rclone.
- **Markerss** — TUI RSS reader (Rust, ratatui): Markdown export, URL bookmarks.
- **Lichtung** — Hugo theme (Go templates / SCSS) powering two personal sites.
- **mumble** — message board (JavaScript, Cloudflare Worker); standalone or embedded pages.

PAPERS & TALKS

Published

Sarkisov Program for Algebraically Integrable Adjoint Foliated Structures 2026
Yifei Chen, Jihao Liu, and Yanze Wang

- **Journal:** *International Mathematics Research Notices*, 2026(6): rnag045.

Flop between algebraically integrable foliations on potentially KLT varieties 2025
Yifei Chen, Jihao Liu, and Yanze Wang

- **Journal:** *International Journal of Mathematics*, 36(11): 2550035.

A Note on the Sarkisov Program 2025
Yifei Chen and Yanze Wang

- **In:** *Higher Dimensional Algebraic Geometry: A Volume in Honor of V. V. Shokurov*, LMS Lecture Note Series, pp. 231–263, Cambridge University Press.

Preprints

All AI-generated, human-verified.

A klt generalized pair with infinitely generated canonical ring 2026
Jihao Liu and Yanze Wang

- arXiv:2608.03258.

Twelve common flex lines in a general pencil of cubics 2026
Jihao Liu and Yanze Wang

- arXiv:2607.26396.

Talks

Sarkisov Program for Algebraically Integrable and Threefold Foliations 2026
Speaker *Xi'an Jiaotong-Liverpool University*